

CHAPTER NINETEEN
**TOMORROW
NEVER KNOWS**



In the history of the Soviet space and missile programs, three singular events stand out as defining moments: the birth of the effort in 1946, the death of Korolev in 1966, and the end of the NI-L3 program in 1974. History, of course, does not separate itself into neat little segments of time, but it would be difficult to find a moment so cataclysmic in the U.S. space program as the Soviet events of 1974. In essence, the year divided the old with the new and a lack of vision with clarity. Completely unknown to the West until the late 1980s, the changes in 1974 were effectively a watershed moment that closed the door on Korolev's determined journey, begun in 1946. What happened after 1974 warrants particular attention, not only as a matter of historical interest, but because the nature of the Soviet piloted space program changed in ways that would have been difficult to foresee at the time of NPO Energiya's formation. Having trudged through failure after failure in the late 1960s and early 1970s, the Soviet Union finally made its arrival as a formidable space superpower in the late 1980s—a full two decades after its only competitor had done the same.

The Rise and Fall of a Space Power

Glushko's ascendance to power at the top of the pyramid coincided with a dramatic shift in fortunes for the Soviet piloted space program. All the failures and catastrophes of 1971 through 1973, especially in the space station effort, seem to have exorcised the demons of the Soviet space program. In 1975, NPO Energiya performed its first fully successful space station mission on *Salyut 4*, one of the two DOS vehicles readied under Mishin. The other one, launched in September 1977 as *Salyut 6*, would finally put the Soviet space program on the slow but persistent track to success. The station's mission was one of the finest success stories in the Soviet space program. In the four years after launch, it hosted sixteen crews, four of which set absolute endurance records for time in space, significantly exceeding the eighty-four-day record set by NASA's Skylab 4 crew during 1973–74. NPO Energiya also introduced two new spacecraft: the Progress, an automated tanker and supply ship, and the Soyuz T, an advanced version of the Soyuz. Ironically, both programs had been initiated by Mishin. It was not simply a matter of setting records but of remarkable maturity in operations. Engineers perfected the very first refueling operations in space, mastered the logistics of having two ships dock to the same station, directed complex repair spacewalks outside the station, managed real-time solutions to contingencies in space, and accumulated a wealth of ground-breaking information on the effects of microgravity on the human organism. The Soviets also extracted maximum political gain from the mission of *Salyut 6* by sending "guest-cosmonauts" from other socialist countries on "friendly" visits. There were no fatalities in the program. It was a stunning return to

In
le
cs
is

he
of
o-
ry
so
of
nt
ral
ns
ay
he
H.
ice
nd
ok
ipt
u-
ish

ind
His
ka,
the
gh,

pas
yuz
aut
95.
lth
to
USA

form, prompting many Western observers to conclude that the Soviets were "ahead" in space. During the same period, the United States accomplished only one piloted flight.

The string of successes in the space station program continued with the operation of *Salyut 7* during the 1982–86 period, culminating with the launch of *Mir* ("World") in February 1986. Crews began visits to *Mir* almost immediately after its launch. In September 1989, two cosmonauts, Viktorenko and Serebrov, began a historic run of ten years of continuous crewed operations. Through the dissolution of the Union of Soviet Socialist Republics in late 1991, *Mir* remained occupied. In 1994–95, Valeriy A. Polyakov, a doctor from the Institute for Biomedical Problems, set the world's endurance record for continuous time spent in space: 438 days.

What had been a closed and secret program began to open up during the early 1990s. *Mir* played a central role in cooperative agreements with Western nations. As part of an arrangement between the United States and the Russian Federation, NASA astronauts began visiting the *Mir* space station in 1995. Seven NASA astronauts, beginning with Norman E. Thagard, spent approximately two and a half years aboard the *Mir* space station between 1995 and 1998. Their quarters were living, breathing, orbiting artifacts of the amazing history of the Soviet space program. The main *Mir* hull is almost identical to the original DOS vehicle that was designed and launched as the first *Salyut* in 1971. The same triumvirate that had built the original *Salyut* created the newer station, but these organizations exist now with different names: RKK Energiya, the *Salyut* Design Bureau, and the Khrunichev State Space Scientific-Production Center.¹ The primary four *Mir* modules—*Kristall*, *Kvant-2*, *Spektr*, and *Priroda*—were all based on the design of the Transport-Supply Ship's main hull, itself part of Chelomey's conception of the Almaz space station complex proposed in the late 1960s. The launch vehicle for *Mir* and its modules is the Proton—a rocket originally known as the UR-500K, proposed by Chelomey as an ICBM in 1960. The delivery vehicles for the complex are the Soyuz TM and the Progress M spacecraft, both derived from Korolev's beloved 7K-OK Soyuz spacecraft, designed in the early 1960s.

Mir, with all its historical significance, was planned for deorbiting by the end of this century. By that time, there will be a more impressive sight in Earth orbit, the International Space Station, a cooperative project involving sixteen countries. As the primary participants, the United States and the Russian Federation will provide most of the materials for this largest ever joint program in the history of space exploration.² The first component of the station, the Zarya Functional Cargo Block, was launched in November 1998 on a Proton booster. The station will be supplied by various modifications of the Soyuz spacecraft. *Mir* operations will probably cease once activities on the International Space Station commence. That singular event will probably mean the end of an independent Russian piloted space program—the end of the journey that Yuriy Alekseyevich Gagarin began in 1961. It will be the beginning of a new and perhaps more exciting voyage.

The *Salyut* and *Mir* space station programs were the most publicized components of the Soviet space program in the 1980s, but they were not, in fact, the most important. The lion's share of the Soviet space budget during the 1980s was taken by the Energiya-Buran effort, the most expensive program in the history of the Soviet space program. After years of delays and cost overruns, NPO Energiya finally launched the first Energiya booster in May 1987. It was the first successful Soviet rocket comparable in power and performance characteristics to NASA's long-defunct Saturn V giant. It was also the first time that the Soviets fired a high-performance LOX-liquid hydrogen rocket engine in operational conditions. What little joy there may have been in such a test was tempered by history. All of the pleas by Korolev and Mishin during the 1960s to develop such engines had fallen on deaf years, leaving Soviet rocket capabilities far behind that

of the United States. It finally took Glushko's change-of-heart about cryogenic propellants before Korolev's dream became a reality. The Energiya booster was fired a second time, in November 1988, when it launched the Soviet space shuttle Buran on a highly impressive fully automated orbital flight. After decades of trying to build a spaceplane, Buran turned out to be the only such Soviet vehicle that ever made it into orbit. It was only fitting that much of the success of Buran benefited from the intensive testing of the small-scale BOR spaceplanes in the 1980s—vehicles that were left over from the ambitious Spiral project from the 1960s.

Despite early expectations of a vigorously expanding Soviet space program, the inevitable disenchantment crept in. As the Soviet economy began to implode, an increasingly free press became the forum for rising criticism of the Energiya-Buran program. By 1993, the effort was in near shambles, with ground models of the Energiya and the Buran rotting away in various plants. In May 1993, the project's Council of Chief Designers requested a final decision from the Russian government.³ The project was formally shelved after seventeen years and 14 billion rubles. For the second time, thousands of Soviet space engineers saw their handiwork disappear into rubble. Many of those who witnessed the demise of the Energiya-Buran project were the same ones who had watched in silence at the abrupt termination of the NI-L3 program. Both projects had their own complex *raison d'être* and their own reasons for fall from grace, but both had one thing in common: they never fulfilled their original promise. The two projects together span the entire period of the piloted space program of the former Soviet Union. For those looking at waste of technology, of knowledge, of money, and ultimately of people, during the postwar Communist era, they need look no further than the NI-L3 and the Energiya-Buran programs.

The End of a Generation

Some would say that Vladimir Nikolayev Chelomey had a career worthy of a great Russian tragedy. After the cancellation of the NI-L3 program, his star seemed to rise for a brief period. In June 1974, he was elected as one of the approximately 1,500 deputies of the Supreme Soviet, the USSR's rubber-stamp parliament. While the legislature had no independent power in the country, membership usually indicated national prominence. In fact, Western observers scouring through the lists of the Supreme Soviet, upon finding Chelomey's name, believed that he was the "new head" of the Soviet space program, a "job previously held by . . . Yangel."⁴ For perhaps a couple of years, he may have also resurrected his ambitious UR-700M Mars landing project. He continued work on the Almaz military space station, two of which were launched between 1974 and 1976 as *Salyut 3* and *Salyut 5*, respectively. He was evidently planning for a major expansion of activities at his design empire, planning much larger versions of Almaz stations serviced by the new Transport-Supply Ship. He even returned to one of his lifelong dreams—the development of an orbital spaceplane.⁵

3. *Utro*, Moscow Ostankino Television, First Channel and Orbita Networks, Moscow, May 25, 1993, 1845 GMT; S. Shamsutdinov, "First Flight of Buran With Tourists on Board Will Take Place on April 12, 1994" (English title), *Novosti kosmonavtiki* 21 (October 9–22, 1993): 40–45.

4. Theodore Shabad, "Russians Indicate Rocket Specialist Heads Space Effort," *New York Times*, July 14, 1974, p. 6.

5. For the "resumption" of the UR-700M program, see Christian Lardier, *L'Astronautique Soviétique* (Paris: Armand Colin, 1992), p. 252. For the Almaz program, see Vladimir Polyachenko, "The 'Pep' of Almaz" (English title), *Krylya rodiny* no. 4 (April 1992): 30–32; Olaf Przybicki, *Almaz: Das supergeheime militärische Orbitalstationsprogramm der UdSSR* (Dresden, Ger.: Institut für Luftfahrt, 1994). For advanced Almaz projects, see I. B. Afanasyev, "Unknown Spacecraft (From the History of the Soviet Space Program)" (English title), *Nauka i zhizn: Nauka, tekhnika, sovetskie kos-*

All of this simply proved too good to be true. In early 1976, one of Chelomey's chief sponsors, Minister of Defense Grechko, succumbed to a heart attack. Chelomey's opponents—primarily Glushko sponsors Ustinov and Kirilenko—reacted immediately. A few weeks after Grechko's death, they bestowed Glushko with an unprecedented honor that hitherto no designer in the space sector had ever held: membership in the Central Committee of the Communist Party. Glushko was officially "elected" at the 25th Communist Party Congress in 1976.⁶ As one observer noted: "From this moment onward, Glushko concentrated in his hands not only the power of an enormous space empire, but also the political power of a commissar, capable of overwhelming anyone in the space establishment."⁷ Glushko's first move was to deny Chelomey any role in the space station program. By 1978, to Chelomey's great alarm, the piloted portion of the Almaz program was terminated. Chelomey had no help from the Ministry of Defense, his usual supporters. They were of the opinion that piloted orbital platforms were less efficient for overhead reconnaissance than automated satellites.⁸

The news just got worse for Chelomey. In 1976, Ustinov, as the new Minister of Defense, took it upon himself to complete the job he had set out to do more than a decade before. Ustinov:

methodically started to strangle Chelomey. He annulled all the military contracts given to Chelomey's enterprise for space flights; he canceled even those that were scheduled in unmanned mode and originally requested by the military.⁹

Perhaps the biggest blow to the Chelomey empire came on June 30, 1981, when Ustinov and Kirilenko pushed through an order that severed Chelomey's important Fili Branch from the main organization and instead made it a branch of NPO Energiya.¹⁰ Given that Chelomey had farmed almost all the key projects to this branch, then known as the Salyut Design Bureau (KB Salyut), he lost all his space- and missile-related projects in one fell swoop. Finally, on December 19, 1981, the Central Committee and the Council of Ministers issued a decree formally terminating not only all work on the Almaz program, but also forbidding Chelomey from any further involvement in the Soviet space program. The official reason for the decision was to "concentrate forces on the creation of the 'Buran' space system."¹¹

Brought to his knees by Ustinov, Chelomey quietly continued to develop naval cruise missiles for the armed forces, which was the original profile of his organization in the 1950s.

6. Two other chief designers in the defense industry were also elected to the Central Committee in 1976: P. D. Grushin from MKB Fakel (air defense and anti-ballistic missiles) and V. F. Utkin from KB Yuzhnoye (ICBMs, spacecraft, and launch vehicles). Grushin had been the first chief designer in the defense industry accorded this honor, with his elections in 1966 and 1971. See Julian Cooper, "The Defense Industry and Civil-Military Relations," in Timothy J. Colton and Thane Gustafson, eds., *Soldiers and the Soviet State: Civil-Military Relations from Brezhnev to Gorbachev* (Princeton, NJ: Princeton University Press, 1990), p. 168.

7. Roald Z. Sagdeev, *The Making of a Soviet Scientist: My Adventures in Nuclear Fusion and Space from Stalin to Star Wars* (New York: John Wiley & Sons, 1993), p. 209.

8. Maxim V. Tarasenko, "The U.S. and Soviet Space Systems Developments as Driven by the Cold War Competition," presented at the 45th Congress of the International Astronautical Federation, IAA-94-IAA.2.2.622, Jerusalem, Israel, October 9–14, 1994.

9. Sagdeev, *The Making of a Soviet Scientist*, pp. 209–10.

10. Yu. P. Semenov, ed., *Raketno-Kosmicheskaya Korporatsiya "Energiya" imeni S. P. Koroleva* (Korolev: RKK Energiya, named after S. P. Korolev, 1996), p. 643; Nina Chugunova, "Chelomey's Cosmonauts: Why There Are No Crews From NPO Mashinostroyeniya in Outer Space" (English title), *Ogonek* 4–5 (January 1993): 24–29.

By all accounts, he never really gave up on his dreams of an ambitious space program, proposing various strategic defense programs throughout the early 1980s. He was not kind to Korolev's memory. In an interview with a journalist late in his life, Chelomey was blunt:

Well, what can I tell you about Korolev? Korolev was a man with a limited education. But he commanded a remarkable technical intuition and was enormously talented as an organizer. Yes. But he couldn't perform even a simple mathematical operation with integrals. He took the circumlunar [program] away from me and then he didn't do it himself. You call that talent?¹²

Ejected from the Soviet space program, Chelomey's will and his reach for success never diminished to his last days. In early December 1984, still in lively health, he was at his dacha, getting ready to go somewhere in his Mercedes. Leaving the car running, he walked out to open the garage door, but the car, still running, moved by itself and pinned his legs against the gate. He was admitted to the hospital with a simple fracture. While in the hospital, he learned that his nemesis Ustinov had suffered a massive heart attack, was paralyzed, and could not speak. Chelomey could be forgiven for believing that his fortunes were about to improve. On the third day in the hospital, on the early morning of December 8, he was speaking to his wife on the telephone when the conversation suddenly stopped. She desperately called the hospital staff, who rushed to his room to find him dead. Doctors suspected a sudden fatal stroke apparently caused by the broken leg. He was seventy years old at the time of his death. Legend has it that Ustinov was brought a piece of paper with a handwritten message stating "Chelomey just died." Ustinov read it and closed his eyes in satisfaction. The first name on the list of signatories of Chelomey's obituary was that of Ustinov.¹³

Today, Chelomey's former organization is called the Scientific-Production Association for Machine Building (NPO Mashinostroyeniya) and is still located at its old grounds at Reutov outside Moscow. Having relinquished hold of its Moscow Branch in 1981, it has little connection to the Russian space program. Its current General Designer, Gerbert A. Yefremov, who succeeded Chelomey, continues to focus mostly on naval cruise missiles. Its only major space-related project is a continuation of the Almaz program—a robotic remote-sensing platform for Earth resources surveying. Three such spacecraft were launched—in 1986, 1987, and 1991—but despite Yefremov's best efforts, funding for a fourth is on a shoestring budget. By September 1994, the organization was in a severe financial crisis, planning to lay off thousands of employees.¹⁴ While the organization may have been in dire straits, Chelomey's legacy, in some ways, remains much more visible than even that of Korolev. Given that Chelomey had his Fili Branch produce most of his space work, the thriving nature of that branch has maintained Chelomey's long shadow across the current Russian space program. The Proton rocket, the *Mir* space station (derived as it was from the original Almaz design), and the *Mir* modules (such as *Spektr* and *Priroda*) all attest to a vision that has remained intact despite the best intentions of Ustinov or Glushko. If Chelomey were alive today, he might have some comfort in knowing that the first

12. Yaroslav Golovanov, *Korolev: fakty i mify* (Moscow: Nauka, 1994), p. 724.

13. *Ibid.*, p. 729; Valeriy Rodikov, "Academician Chelomey and His Times" (English title), in V. Shcherbakov, ed., *Zagadki zvezdnykh ostrovov: kniga pyataya* (Moscow: Molodaya gvardiya, 1989), pp. 35–36; "Vladimir N. Chelomei, Soviet Rocket Scientist," *New York Times*, December 15, 1984, p. 28; Christian Lardier, "Soviet Space Designers When They Were Secrets," presented at the 47th Congress of the International Astronautical Federation, IAA-96-IAA.2.2.00, Beijing, China, October 7–11, 1996.

element of the International Space Station, the Zarya Functional Cargo Block, is based on the design of the service module of the Transport-Supply Ship—a program that he pushed into approval in 1970. Zarya was designed, built, and delivered to NASA by the Khrunichev State Space Scientific-Production Center, a conglomerate of Chelomey's former Fili Branch (now called the Salyut Design Bureau) and the Khrunichev Machine Building Plant, established by governmental order on June 7, 1993.¹⁵

Chelomey's nemesis Ustinov had a meteoric career. With the exception of Korolev, Ustinov may have been the single most important individual in the emergence of the Soviet space program. At the same time, he is probably also the most overlooked. Scarcely mentioned in the Western historiography of the Soviet space program, Ustinov was at the center of the vortex of events of the Soviet space effort from 1946 to 1984, close to a forty-year span of time. Even in Russia, there have been no biographies of the man, nor is their evidence to suggest that he left personal memoirs. Of course, Ustinov's importance was not limited to the space program. He directly oversaw the tremendous growth and arrival of the Soviet Union as a formidable military player in world politics. Between 1965 and 1976, Ustinov was the Secretary of the Central Committee for Space and Defense, but he did not achieve his lifelong dream of entering the ranks of the Politburo until April 1976 with his appointment as the first Soviet civilian to serve as the Minister of Defense. His tenure at the post was a time fraught with unprecedented tensions with the United States, particularly over the Soviet invasion of Afghanistan and the Americans' stationing of Pershing missiles in Western Europe. He was reportedly in ill health in the early 1980s and was fast becoming a victim of Communist Party politics.¹⁶ Ustinov died on December 20, 1984, after a two-month illness at the age of seventy-six.¹⁷ In one of the bitterest of all ironies, his death came just twelve days after Chelomey had passed away. In an indication of new times, his death was first announced by Mikhail S. Gorbachev, considered at the time a fast-rising personality in the upper of echelons of power. Not surprisingly, Westerners writing about his life's achievements at the time almost completely missed his contribution to the creation and sustenance of the Soviet space program.¹⁸

Among the other heavy hitters of the Soviet space program, Leonid V. Smirnov, the former Chairman of the Military-Industrial Commission, served in that position from March 1963 to December 1985, managing the development and creation of several new generations of Soviet weaponry. During the reshuffling after Ustinov's death, Smirnov retired—the last of the powerful defense industry juggernauts who built up the military might of the USSR. Although still alive at the time of this writing, the eighty-three-year-old Smirnov has remained completely out of the public eye. His personal reminiscences would no doubt be a priceless asset to understanding Soviet motives during the Cold War.

Sergey A. Afanasyev, the Minister of General Machine Building—that is, the "space and missile" ministry,—served in that capacity from March 1965. After the death in 1976 of his chief sponsor, Minister of Defense Grechko, Afanasyev's star dropped rapidly. In April 1983, Ustinov finally had him fired. He was given the far less important job of Minister of Heavy and Transport Machine Building, which was a sector outside the defense industry. With his ambition of one day entering the ranks of the Politburo crushed, Afanasyev trudged through his new dreary job, before finally being forced to retire prematurely in July 1987.¹⁹ The "Big Hammer,"

as he was nicknamed by many in the space industry, has retained contacts with the Russian space industry as a "Chief Scientific Consultant to the General Designer" of RKK Energiya. In one of his very rare published memoirs of the space era, he had only favorable words to say of Ustinov, despite the obvious clashes between the two men.²⁰ At the time of this writing, he was eighty-one years old.

The Soviet space program had originated as an arm of the artillery sector of the Soviet armed forces, and as such, there were a number of important artillery officers who played prominent roles in guiding the entire effort. There was probably no one officer more important than Lt. General Georgiy A. Tyulin, whose involvement in the Soviet missile program began in 1944, when he was a young lieutenant charged with assessing German rocket technology. In March 1965, he was appointed Afanasyev's First Deputy in the Ministry of General Machine Building. During the 1960s, he served as the chair of various State Commissions, including those for the later Vostok missions, the Voskhod program, the L1 circumlunar project, and various lunar and interplanetary probes. He remained at his ministerial post until 1976, when, rumor has it, Afanasyev fired him for being part of the "Ustinov camp."²¹ Forced into retirement, the quiet and reticent Tyulin returned to teach theoretical mechanics at the M. V. Lomonosov Moscow State University.²² In 1987, he began writing publicly about his deep well of experience in the missile and space programs—articles that have been remarkably valuable in filling in the gaps of this secret history. After a long illness, he died in April 1990 at the age of seventy-five.²³

Tyulin was certainly better known than Vasilii M. Ryabikov, who chaired the State Commission for Sputnik. One of the most mysterious figures in the early Soviet space program, Ryabikov was instrumental in the process of approving the first Sputnik launch. His early career was under Ustinov's shadow, but for a brief period in the 1950s, he emerged as one of the power players in the defense industry, only to disappear into relative oblivion. Almost nothing is known about his personal history. After his "ejection" from the defense industry, he served as the First Deputy Chairman of the State Planning Organ (better known as "Gosplan") until his death on July 19, 1974, at the age of sixty-seven. Even in recent years, Russian historians have generally shied away from any in-depth analysis of Ryabikov's role in the genesis of Sputnik. It is a curious omission for a man who may have facilitated the inauguration of the space era in 1957.

Of the two other major artillery officers from the space era, one remains alive. Lt. General Kerim A. Kerimov was demoted out of his ministry position in 1974, but remained the chair of the State Commission for Soyuz until 1991, a position he had assumed in 1966. He oversaw the launch of every single Soyuz spacecraft to the *Salyut* and *Mir* space stations during that period.²⁴ At the time of his retirement, he was officially the First Deputy Director of the Central Scientific-Research Institute for Machine Building (TsNIMash), the leading research and development institution in the Soviet space industry.²⁵ At this writing, he was eighty-two years old. Lt. General Yuriy A. Mozhzhorin, the powerful Director of TsNIMash, remained in that post until December 1990, completing almost thirty years of service as one of the primary policymakers in the Soviet space program. He continued to be active in chronicling the history of the Soviet missile and space programs and served as editor of the series of memoirs titled *Dorogi v*

15. Zhiltsov, ed., *Gosudarstvennyy kosmicheskiy*, pp. 126–27.

16. Sagdeev, *The Making of a Soviet Scientist*, pp. 258–59.

17. Serge Schmemmann, "Defense Minister of Soviet Union Is Dead at Age 76," *New York Times*, December 22, 1984, pp. 1, 6.

20. See Yu. A. Mozhzhorin, et al., eds., *Dorogi v kosmos: I* (Moscow: MAI, 1992), pp. 34–48.

21. Sagdeev, *The Making of a Soviet Scientist*, p. 201.

22. Col. M. Rebrov, "Where the Cranes Fly" (English title), *Krasnaya zvezda*, September 19, 1987, pp. 3–4.

23. "G. A. Tyulin," *Krasnaya zvezda*, April 25, 1990, p. 4.

element of the International Space Station, the Zarya Functional Cargo Block, is based on the design of the service module of the Transport-Supply Ship—a program that he pushed into approval in 1970. Zarya was designed, built, and delivered to NASA by the Khrunichev State Space Scientific-Production Center, a conglomerate of Chelomey's former Fili Branch (now called the Salyut Design Bureau) and the Khrunichev Machine Building Plant, established by governmental order on June 7, 1993.¹⁵

Chelomey's nemesis Ustinov had a meteoric career. With the exception of Korolev, Ustinov may have been the single most important individual in the emergence of the Soviet space program. At the same time, he is probably also the most overlooked. Scarcely mentioned in the Western historiography of the Soviet space program, Ustinov was at the center of the vortex of events of the Soviet space effort from 1946 to 1984, close to a forty-year span of time. Even in Russia, there have been no biographies of the man, nor is their evidence to suggest that he left personal memoirs. Of course, Ustinov's importance was not limited to the space program. He directly oversaw the tremendous growth and arrival of the Soviet Union as a formidable military player in world politics. Between 1965 and 1976, Ustinov was the Secretary of the Central Committee for Space and Defense, but he did not achieve his lifelong dream of entering the ranks of the Politburo until April 1976 with his appointment as the first Soviet civilian to serve as the Minister of Defense. His tenure at the post was a time fraught with unprecedented tensions with the United States, particularly over the Soviet invasion of Afghanistan and the Americans' stationing of Pershing missiles in Western Europe. He was reportedly in ill health in the early 1980s and was fast becoming a victim of Communist Party politics.¹⁶ Ustinov died on December 20, 1984, after a two-month illness at the age of seventy-six.¹⁷ In one of the bitterest of all ironies, his death came just twelve days after Chelomey had passed away. In an indication of new times, his death was first announced by Mikhail S. Gorbachev, considered at the time a fast-rising personality in the upper of echelons of power. Not surprisingly, Westerners writing about his life's achievements at the time almost completely missed his contribution to the creation and sustenance of the Soviet space program.¹⁸

Among the other heavy hitters of the Soviet space program, Leonid V. Smirnov, the former Chairman of the Military-Industrial Commission, served in that position from March 1963 to December 1985, managing the development and creation of several new generations of Soviet weaponry. During the reshuffling after Ustinov's death, Smirnov retired—the last of the powerful defense industry juggernauts who built up the military might of the USSR. Although still alive at the time of this writing, the eighty-three-year-old Smirnov has remained completely out of the public eye. His personal reminiscences would no doubt be a priceless asset to understanding Soviet motives during the Cold War.

Sergey A. Afanasyev, the Minister of General Machine Building—that is, the "space and missile" ministry,—served in that capacity from March 1965. After the death in 1976 of his chief sponsor, Minister of Defense Grechko, Afanasyev's star dropped rapidly. In April 1983, Ustinov finally had him fired. He was given the far less important job of Minister of Heavy and Transport Machine Building, which was a sector outside the defense industry. With his ambition of one day entering the ranks of the Politburo crushed, Afanasyev trudged through his new dreary job, before finally being forced to retire prematurely in July 1987.¹⁹ The "Big Hammer,"

15. Zhiltsov, ed., *Gosudarstvennyy kosmicheskiy*, pp. 126–27.

16. Sagdeev, *The Making of a Soviet Scientist*, pp. 258–59.

17. Serge Schmemmann, "Defense Minister of Soviet Union Is Dead at Age 76," *New York Times*, December 22, 1984, p. 1, 6.

as he was nicknamed by many in the space industry, has retained contacts with the Russian space industry as a "Chief Scientific Consultant to the General Designer" of RKK Energiya. In one of his very rare published memoirs of the space era, he had only favorable words to say of Ustinov, despite the obvious clashes between the two men.²⁰ At the time of this writing, he was eighty-one years old.

The Soviet space program had originated as an arm of the artillery sector of the Soviet armed forces, and as such, there were a number of important artillery officers who played prominent roles in guiding the entire effort. There was probably no one officer more important than Lt. General Georgiy A. Tyulin, whose involvement in the Soviet missile program began in 1944, when he was a young lieutenant charged with assessing German rocket technology. In March 1965, he was appointed Afanasyev's First Deputy in the Ministry of General Machine Building. During the 1960s, he served as the chair of various State Commissions, including those for the later Vostok missions, the Voskhod program, the L1 circumlunar project, and various lunar and interplanetary probes. He remained at his ministerial post until 1976, when, rumor has it, Afanasyev fired him for being part of the "Ustinov camp."²¹ Forced into retirement, the quiet and reticent Tyulin returned to teach theoretical mechanics at the M. V. Lomonosov Moscow State University.²² In 1987, he began writing publicly about his deep well of experience in the missile and space programs—articles that have been remarkably valuable in filling in the gaps of this secret history. After a long illness, he died in April 1990 at the age of seventy-five.²³

Tyulin was certainly better known than Vasilii M. Ryabikov, who chaired the State Commission for Sputnik. One of the most mysterious figures in the early Soviet space program, Ryabikov was instrumental in the process of approving the first Sputnik launch. His early career was under Ustinov's shadow, but for a brief period in the 1950s, he emerged as one of the power players in the defense industry, only to disappear into relative oblivion. Almost nothing is known about his personal history. After his "ejection" from the defense industry, he served as the First Deputy Chairman of the State Planning Organ (better known as "Gosplan") until his death on July 19, 1974, at the age of sixty-seven. Even in recent years, Russian historians have generally shied away from any in-depth analysis of Ryabikov's role in the genesis of Sputnik. It is a curious omission for a man who may have facilitated the inauguration of the space era in 1957.

Of the two other major artillery officers from the space era, one remains alive. Lt. General Kerim A. Kerimov was demoted out of his ministry position in 1974, but remained the chair of the State Commission for Soyuz until 1991, a position he had assumed in 1966. He oversaw the launch of every single Soyuz spacecraft to the *Salyut* and *Mir* space stations during that period.²⁴ At the time of his retirement, he was officially the First Deputy Director of the Central Scientific-Research Institute for Machine Building (TsNII Mash), the leading research and development institution in the Soviet space industry.²⁵ At this writing, he was eighty-two years old. Lt. General Yuriy A. Mozzhorin, the powerful Director of TsNII Mash, remained in that post until December 1990, completing almost thirty years of service as one of the primary policymakers in the Soviet space program. He continued to be active in chronicling the history of the Soviet missile and space programs and served as editor of the series of memoirs titled *Dorogi v*

20. See Yu. A. Mozzhorin, et al., eds., *Dorogi v kosmos: I* (Moscow: MAI, 1992), pp. 34–48.

21. Sagdeev, *The Making of a Soviet Scientist*, p. 201.

22. Col. M. Rebrov, "Where the Cranes Fly" (English title), *Krasnaya zvezda*, September 19, 1987, pp. 3–4.

23. "G. A. Tyulin," *Krasnaya zvezda*, April 25, 1990, p. 4.

kosmos (Roads to Space), the first volume of which was published in 1992. He died on May 15, 1998, at the age of seventy-seven.

The Designers

All six members of the original Council of Chief Designers are deceased. Korolev, of course, was the first to go in January 1966. Academician Nikolay A. Pilyugin, Korolev's closest friend on the council, died on August 2, 1982, at the age of seventy-four, after a long bout with diabetes.²⁶ His obituary was signed by Brezhnev, Andropov, Gorbachev, and Chernenko, all heads of the Soviet state at various points. Chief Designer Mikhail S. Ryazanskiy died after a long battle with cancer of the prostate gland on August 7, 1987. Academician Viktor I. Kuznetsov passed away four years later on March 22, 1991. The last member (aside from Glushko), Academician Vladimir P. Barmin, lived to the age of eighty-four, heading the organization he had founded until his death on July 17, 1993.²⁷ In one of his last interviews, Barmin waxed philosophical about the constraints of the Communist era:

*... I have been working as a Chief Designer for more than fifty years, and have been "open" to the press only in recent years. My articles in the newspaper Pravda used to be under a pseudonym, Professor Vladimirov. ...*²⁸

Although he was not a member of the original council, General Designer Academician Nikolay D. Kuznetsov, responsible for creating the NI's rocket engines, played a major role in the rise and fall of the huge project. Despite Glushko's order to have all NI-related materials destroyed, Kuznetsov, at his own risk, preserved ninety-four engines of the first, second, third, and fourth stages at the storage facilities of the Trud Scientific-Production Association. All were completely ready for operational use. In addition, he also hid away fifty to sixty experimental units, ready for future developmental work. Kuznetsov's gamble paid off when in the early 1990s, major U.S. aerospace companies expressed interest in using the engines for the next generation of expendable U.S. launch vehicles. In late 1993, the Aerojet Propulsion Division imported a flight-ready version of the NK-33, believing the design to be of "very modern technology compared with what the U.S. has in LOX/kerosene engines."²⁹ In 1995, Kuznetsov's organization went head-to-head with Glushko's firm bidding for their respective engines on new versions of the Atlas or Delta rockets. Although Glushko's engines won that bid, the NI engines may still see the light of day.³⁰ In 1996–97, Kistler Aerospace Corporation of Kirkland, Washington, signed an agreement with Kuznetsov's former organization to use the NI's NK-33 and NK-43 engines on the company's K-1 reusable launch vehicle. In what could be a fitting legacy of the NI rocket, the first K-1 vehicle is expected to use the very same engine units that were meant for use on the canceled 8L launch of the NI in 1974.³¹ Sadly, Kuznetsov himself will not be witness to their use. At the age of eighty-four, he died on July 31, 1995.³²

26. "Academician N. A. Pilyugin," *Pravda*, August 3, 1982, p. 3; "Nikolai A. Pilyugin, 74, Dies; Was Key Soviet Space Figure," *New York Times*, August 3, 1982, p. A19.

27. "Academician V. P. Barmin," *Krasnaya zvezda*, July 22, 1993, p. 4.

28. V. Smirnov, "Topical Interview: Space Starts With the People on the Ground" (English title), *Aviatsiya i kosmonavtika* no. 10 (October 1992): 2–3.

29. Michael A. Dornheim, "Aerojet Imports Trud NK-33 Engine," *Aviation Week & Space Technology*, October 25, 1993, p. 29.

30. Dennis Newkirk, *Russian Space Review 1996* (Roselle, IL: Dennis Newkirk, 1997).

31. V. S. Anisimov, T. C. Lacey, and J. Andrews, "Evolution of the NK-33 and NK-43 Reusable LOX/Kerosene Engines," presented at the 22nd AIAA/ASME/ASCE Joint Propulsion Conference & Exhibit

The legacy of the NI also survives in the high-performance LOX–liquid hydrogen engines that were developed and tested in the early 1970s but were never used in flight. Most notable was Chief Designer Lyulka's IID57 engine. The engine's production stopped in 1975 after 105 were built. During the testing period, the engine had accumulated more than 53,000 seconds of full-engine run time. In late 1993, Aerojet expressed interest in using the engine for its single-stage-to-orbit program.³³ Chief Designer Isayev's IID56, another LOX–liquid hydrogen engine developed for the NI, became the center of controversy in 1993, when the sale of the engine to the Indian Space Research Organization was blocked by the U.S. government, which had concerns over their potential application in military missile systems.³⁴ After further negotiations, the Russian Federation delivered the first such engine to India in September 1998.

Although their names have not been prominent in Western histories of the Soviet space program, a number of men from the old Korolev design bureau played very critical roles in the road that led to Sputnik. Certainly from an engineering standpoint, there was no other individual more important in the genesis of Sputnik than Mikhail K. Tikhonravov. Overshadowed by the much more famous Korolev, Tikhonravov's role in the early space program was quite likely as important as that of his boss. With his landmark 1954 report on artificial satellites, he set off a process that ended with the launch of Sputnik in 1957. After Sputnik, Tikhonravov led the teams that designed the first piloted spacecraft, the first automated lunar probes, and the first Soviet reconnaissance satellites. He also contributed to policy by co-authoring important long-range plans for Korolev's design bureau. He continued work under Korolev, vigorously supporting piloted space exploration against those who believed in robotic exploration. He seems to have retired from the design bureau after Korolev's death and returned to teaching and writing. He died on March 4, 1974, at the age of seventy-four, after a spectacular career that had begun with his design of the first Soviet liquid-propellant rocket, the "09" in 1933. As with many other important individuals in Soviet space history, his life and his remarkable contributions remain drowned out by the flood of writings on Korolev. As a mark of respect to his memory, in February 1995, the Russian Military Space Forces renamed their leading space research institute, TsNII-50, after Tikhonravov.³⁵



Mikhail Tikhonravov was one of the most important engineers behind the emergence of the Soviet space program. He designed the first Soviet liquid-propellant rocket (in the 1930s), performed research to optimize early ICBM designs (in the 1940s and 1950s), and was the leading engineer behind the genesis of Sputnik. In later years, he was also instrumental in the design of the Vostok spaceship. (files of Peter Gorin)

33. "Aerojet, Lyulka Push D-57 for SSTO Validation," *Aviation Week & Space Technology*, October 11, 1993, pp. 50–51.

34. Jeffrey L. Lenorovitz, "Space Systems/Loral Books Proton Launches," *Aviation Week & Space Technology*, September 20, 1993, pp. 90–91.

With respect to the Korolev "high guard"—his key deputies—most have passed away. The *de facto* head of all piloted space programs at OKB-1 throughout the 1960s and 1970s, Konstantin D. Bushuyev, lived to serve as the director of the Soviet side of the Apollo-Soyuz Test Project in 1975. Although his true position, a Chief Designer at NPO Energiya, was kept tightly under wraps, he told his U.S. counterparts on one occasion that "he had started working with Korolev right after World War II. . . ."36 Officially, during the entire joint project, he was forced to pretend that he was actually an employee of the Institute of Space Research under the USSR Academy of Sciences. This charade played out right up to his sudden death of a heart attack on October 26, 1978, at the age of sixty-four. He was apparently suffering from a toothache and was headed to the hospital when he suddenly dropped dead in a corridor. Unsure of how to facilitate the funeral of a figure in the Soviet space program whose identity was known *before* his death, Soviet officials chose the most ludicrous path. As one observer noted later: "After his death, instead of having a decent funeral at the former Korolev Design Bureau, where he had spent most of his active working life, the final sad ceremony was moved to the [Institute of Space Research], simply as a cover. . . ."37

As for the two "fathers" of the N1, Sergey O. Okhapkin died in March 1980 at age seventy.³⁸ Given a different course of events in 1974, Okhapkin might very well have succeeded Mishin as head of the organization, because he had served as Mishin's First Deputy since 1966. The other N1 designer, Sergey S. Kryukov, remains alive today, and he occasionally writes in the Russian media on the topic. He had one of the more interesting careers of any of Korolev's protégés. A few years after Korolev's death, on March 30, 1970, Kryukov left TsKBEM and joined the Lavochkin Design Bureau as the famous Babakin's First Deputy, thus turning his back on the N1 and piloted spacecraft to focus on robotic probes. After Babakin's death, on August 26, 1971, Kryukov took over the design bureau and guided the organization through a mixed bag of lunar and interplanetary missions. Having become the victim of political maneuvering over a proposed Martian sample return project, Kryukov returned to his original place of work, then NPO Energiya. On November 17, 1977, he was appointed the First Deputy General Designer under Glushko. After overseeing the immensely successful *Salyut 6* space station missions, he retired in January 1982.³⁹ Still a "scientific consultant" to Energiya, Kryukov, at the time of this writing, is eighty-one years old.

Of all of Korolev's deputies, perhaps the most well known is Boris Ye. Chertok. His career started with Mishin and Bushuyev at the famous Bolkhovitinov Design Bureau in the late 1930s. Chertok remained a powerful figure at Energiya through the 1980s, but he never rose to the top of the organization. Although he retired in 1991 from his official duties as Deputy General Designer, he continues to maintain his offices at the giant organization as a "Chief Scientific Consultant." Still full of verve and energy at the age of eighty-seven, Chertok recently admitted that "in the N1-L3 project we . . . made serious mistakes."⁴⁰ He is one of the few men who, having lived through those historic times, has put pen to paper, and he is in the midst of publishing a multiple-volume set of priceless reminiscences. Incredibly detailed and remarkably devoid of partiality, these memoirs, titled *Raketi i lyudi* (*Rockets and Men*), cover everything from Chertok's early forays into Germany in search of A-4 missiles in 1945 to the

demise of the Energiya-Buran system during the early 1990s. Much more accessible than many other old-timers, Chertok continues to travel all over the world, including the United States, to speak of his life. He also has one foot in the future. His current project is a modest system of communications satellites in low-Earth orbit to serve the general populace.⁴¹

From the scientific community, there was probably no one individual who wielded as much influence as Academician Mstislav V. Keldysh, President of the USSR Academy of Sciences from 1961 and one of the most brilliant mathematicians of his generation. Unlike Korolev, Keldysh's personal contributions span the gamut from purely technical to purely managerial. During the 1950s, Keldysh personally participated and directed top-secret studies on the optimal design characteristics of multistage rockets, the question of returning a satellite from Earth orbit, the theory of passive gravitational stabilization of satellites, the calculation of various satellite orbits, and the mathematical analyses of optimal trajectories for flight to the Moon, Mars, and Venus. This research was performed at two institutions, both of which Keldysh headed simultaneously: the Department of Applied Mathematics of the V. A. Steklov Mathematics Institute under the Academy of Sciences and NII-1 under the Ministry of Aviation Industries. At the latter institute, Keldysh also initiated work on high-performance ramjet engines and nuclear rocket engines.

From 1961, after he was appointed to head the Academy of Sciences, Keldysh's most important contributions were as the chair of the Interdepartmental Scientific-Technical Council for Space Research. With Keldysh as chair, various permutations of this council served as "expert commissions" for several dozen different military and space programs.⁴² In 1975, Keldysh stepped down as President of the Academy of Sciences because of ill health. A man with a calm disposition who rarely, if ever, lost his temper, Keldysh's favorite form of relaxation was collecting prints of the Impressionists. He died on June 24, 1978, at the age of sixty-seven, sitting at the wheel of his car in the garage of his country home.⁴³ Keldysh's ashes were interred in the Kremlin Wall, an honor reserved for only the most revered Soviet citizens of this century. All fourteen members of the Politburo signed his obituary. Throughout his extraordinary life, there were probably few sectors of the Soviet military-industrial complex Keldysh did not influence with his scientific contributions or advisory activities.

Glushko

Academician Valentin Petrovich Glushko effectively headed the Soviet space program from 1974 for a fifteen-year period, and during that time, some would argue, there was almost a cult of personality surrounding his name. Glushko, having a hand in the editorial supervision of all books related to space exploration, made sure that his role and contributions to the development of Soviet space technology were placed in a favorable light. If in a 1957 speech at Korolev's fiftieth birthday, Glushko could say "Korolev occupies first place after Tsiolkovskiy" in the development of Soviet rocketry, he did not hesitate in later years to insert his name before Korolev in all histories of the Soviet space program.⁴⁴ But with so much power, Glushko was still unable to carry out one of his most coveted dreams—piloted landing expeditions to the

36. Edward Clinton Ezell and Linda Neumann Ezell, *The Partnership: A History of the Apollo-Soyuz Test Project* (Washington, DC: NASA Special Publication (SP)-4209, 1978), p. 288.

37. Sagdeev, *The Making of a Soviet Scientist*, p. 174.

38. Golovanov, *Korolev*, p. 480.

39. Yu. Markov, *Kurs na Mars* (Moscow: Mashinostroyeniye, 1989), pp. 25–26; Semenov, ed., *Raketo-Kosmicheskaya Korporatsiya*, p. 641. Note that the first source gives a slightly different date, December 1, 1982.

41. *Ibid.*

42. N. Chentsov, "World Famous, But Secret in Every Way" (English title), *Nauka i zhizn* no. 2 (February 1991): 102–07; V. S. Avduyevskiy and M. Ya. Marov, "Mstislav Vsevolodovich Keldysh and Space Research" (English title), *Zemlya i vseennaya* no. 3 (May–June 1991): 46–52.

43. "M. V. Keldysh Dies; Mathematician Led Academy in Soviet," *New York Times*, June 27, 1978.

44. For an interesting analysis of Glushko and his role in Soviet history, see Gerson Margov, "You

Moon and Mars. At various points throughout the 1980s, he continued to bring this idea up to the Soviet leadership, but each time it was rejected. Knee-deep in the Energiya-Buran program, the Soviet military had little interest in funding another repeat of the N1-L3 debacle.⁴⁵

Despite Glushko's remarkable rise to prominence as the reigning emperor of the Soviet space program, he was still a man trapped within his times. Few photographs of him were published, and apart from the cursory details of his professional history, outsiders had no clue about his personal life. Recently, there has been a tendency to paint Glushko as some kind of evil player of the Soviet space program, the man who single-handedly destroyed the N1 program—first when he broke off relations with Korolev in the 1960s and second when he canceled the program in the 1970s. But this revisionism comes perhaps more from haphazard retroactive assessments than any in-depth analysis. While Korolev has been humanized by countless biographies, Glushko still remains an enigma—a man whose only motive, it seems, was to sabotage Korolev's dreams. Is it possible to bring Glushko down to the level of a human, flawed perhaps, but at the end of the day having the same ideals of space exploration as those of Korolev? He was apparently well versed in the finer arts, such as music, painting, and literature, was good at drawing, spoke fluently in five languages, and regularly kept up with non-scientific foreign journals. His deputies remember him as a man who had an "eye for style and flair for detail . . . he would always be elegantly dressed."⁴⁶ He was married several times. Apart from his clearly notable contributions to the space program, Glushko also spent years completing a forty-volume series for the Academy of Sciences on the topics related to rocket propulsion theory. Overall, he published more than 250 scientific works.

In 1989, Mikhail F. Rebrov, a Soviet military journalist acquainted with Glushko, wrote a very candid account of the general designer's life:

He was never weak nor banal—traits that frequently accompany material and professional success. As he himself said, his life was a long, difficult search which essentially consisted of attempting to reach the desired level of simplicity upon mastering incredibly complex designs. He apparently gave himself over fully to his life's main work, and was ready to sacrifice for it. But that was only the way things seemed. Where Korolev could at some point, after judiciously evaluating his capabilities, reserve the main strategic problem for himself, and turn some problems over to his students, Glushko did not let anything out of his hands.⁴⁷

Referring to the final years of his life, Rebrov wrote:

Nothing it seems could quench his thirst for activity, his frenzied passion to go down in history by completing what would come to be called "the first in the world." He was compared to the director of an enormous orchestra who was enchanted by the dream of playing something in such a way that would make the world talk about "the new Russian triumph in space." And, to a certain extent, he succeeded in this. . . .⁴⁸

45. Glushko also tried to generate public interest in his Mars plans by writing in newspapers. See, for example, V. Glushko, Yu. Semenov, and L. Gorshkov, "Fantasy on the Drawing Board: The Road to Mars" (English title), *Prauda*, May 24, 1988, p. 3.

46. Boris Katargin and Leonid Sternin, "Pushing Back the Missile Technology Frontiers," *Aerospace Journal*

The Korolev-Glushko fallout has been discussed much in recent years, but most accounts attribute this fracture to personal vendetta more than professional opinions. It seems more likely, however, that both men were acting perfectly within the bounds of their experience over the N1 propellant issue, with Glushko supporting storables and Korolev cryogenics. Both men had solid reasons for their choices—rationales that had almost nothing to do with personality conflicts or outright hatred. The two had, after all, worked together through the Purges, through prison, and through the Stalinist era and maintained their cooperation. Recent accusations notwithstanding, there is no evidence to suggest that Glushko's testimony led to Korolev's imprisonment in the 1930s. In fact, both men acted with remarkable honor, given the exigencies of the day. Perhaps the tragedy of Glushko's life, if there is one, is that his ultimate ambition of being known as the most important person in the tapestry of the Soviet space program will never come true. He will always be behind Korolev, and he probably knew this fact very well. As early as 1968, a couple of years after Korolev's death, when a journalist asked Glushko about Korolev, Glushko replied, "But why are you always going on about Korolev! Korolev! And what was Korolev? He was just a thin metallic pipe. Inside it I placed my engines, Pilyugin—his instruments. Barmin put it on the launch pad and it flew. . . ."⁴⁹

By the late 1980s, Glushko was seriously ill and partially paralyzed, to the point that a special stamp was made for him because he could not even sign his own name. While he was able to attend the first launch of his life's dream, Energiya, he was too ill to be at the Baykonur Cosmodrome for the launch of Buran. He continued working from his bed, asking for reports from his deputies on every little detail. In August 1988, knowing his days were numbered, he told one of his deputies that he wanted his ashes to be placed in an urn and kept in a safe place so that one day it may be taken to the surface of Venus.⁵⁰ Just fifty-six days after the first and last flight of the Buran space shuttle, on January 10, 1989, Glushko passed away in Moscow at the age of eighty.⁵¹ Even Gorbachev paid his respects. Thus ended the journey that had begun in 1923, when a fifteen-year-old boy had written to Tsiolkovskiy about rockets traveling in space.

Mishin

If Glushko is conventionally known as the man who sabotaged the N1 program, then the popular retrospective evaluations of the contributions of Academician Vasily Pavlovich Mishin have been even less generous. One can almost randomly pick up any article on the history of the Soviet space program, and there will probably be a disparaging remark about Mishin. The hapless Mishin, after all, presided over the most ignominious period in Soviet space history. What better way to explain away all those failures than to attribute it to a short-tempered, impulsive, and unskilled manager who had a drinking problem? In all likelihood, there is probably much truth in the negative assessment of Mishin's role as a chief designer. He made some exceptionally poor decisions and pursued causes that collectively had seriously regressive repercussions on the effort as a whole. But like any figure in a complex history, his contributions were not one dimensional. In fact, quite possibly, his role has been demeaned unfairly.

After he was fired in May 1974, Mishin declined to take up Brezhnev's offer to help find a job, and he returned to full-time teaching at the prestigious Moscow Aviation Institute, his alma mater. He had originally founded the Cosmonautics Faculty at that institute in 1959 and taught

49. Golovanov, *Korolev*, p. 707.

50. Semenov, ed., *Raketo-Kosmicheskaya Korporatsiya*, p. 434; Katargin and Sternin, "Pushing Back the Missile Technology Frontiers." Note that in the latter source, the authors state that Glushko wanted his ashes taken

on a part-time basis for fifteen years. After his dismissal, he went back only once to his old design bureau, as part of a project to document Korolev's scientific heritage. While he was no longer involved in the mainstream Soviet space program, Mishin continued to pursue an academic career focused on space technology. As part of his teaching, he directed a design project that led to the creation of the *Radio-1* and *Radio-2* amateur communications satellites.⁵² His name was, of course, never mentioned in any histories of the period. Glushko apparently wished to remove Mishin from history. Although he was allowed to publish under his own name, Mishin wrote only technical books or contributed to historical works without being able to admit his own personal role in any space or missile project.⁵³

His relatively obscure existence was interrupted dramatically in 1985 when the KGB abruptly summoned him for questioning about his relationship to a journalist named Suslov who had interviewed him. The KGB agents told him that Suslov was under arrest on charges of passing Soviet secrets to the West; they believed that Mishin was his accessory and threatened to strip him of membership in the Communist Party and put him on trial. The KGB finally dropped the case when they could not find evidence to implicate Mishin. He had simply been one of the people Suslov interviewed.⁵⁴

Given that Mishin was not allowed to talk about his role in the history of the Soviet missile and space program, few people in the West were even aware of his significant role. His name was first mentioned by a Soviet émigré in 1982 and later by a French journalist in 1985, but most Western analysts remained unconvinced, believing that it was Yangel or perhaps Chelomey who had succeeded Korolev in 1966.⁵⁵ As the new era of *glasnost* ("openness") dawned on the Soviet Union during the mid-1980s, it slowly opened up the cellar doors of long-forgotten tales. A nation began to rewrite its history. In 1986, journalist Yaroslav K. Golovanov was allowed, by special clearance of the Central Committee, to write on the original group of cosmonauts. In a six-part article in the official Soviet newspaper *Izvestiya*, Golovanov revealed the events behind Gagarin's historic mission.⁵⁶ Among the more tantalizing revelations were the names of all twenty men who had been selected as cosmonauts in 1960. Until then, Soviet censors had allowed the publication of only the twelve who had flown into space.

Unflown cosmonauts were not the only ones who benefited from this free exchange of information. In late 1987, as part of celebrations for the thirtieth anniversary of the launch of the first Sputnik, the Soviet astronomy and space journal *Zemlya i vseennaya* (*Earth and Universe*) published a short article by Mishin in which he openly revealed that he had succeeded Korolev.⁵⁷ *Glasnost* may have meant openness, but Glushko made sure that there was no talk of the piloted lunar program, for to admit such a history was to admit that not only did the USSR race the United States to the Moon, but that it had lost. It finally took Glushko's death in January 1989 to change the climate. In July 1989, a relatively obscure newspaper named *Poisk* (*Search*) published a short article consisting of a few diary entries from the personal journal of Air Force General Nikolay P. Kamanin. There was no ambiguity in his writing: the Soviet Union had had a piloted lunar

52. Jacques Villain, ed., *Baïkonour: la porte des étoiles* (Paris: Armand Colin, 1994), p. 136.

53. Two of his books from the 1970s and 1980s are: V. P. Mishin, *Vvedeniye v mashinnoye proyektirovaniye letatelnykh apparatov* (Moscow: Mashinostroyeniye, 1978), and V. P. Mishin, *Osnovy proyektirovaniya letatelnykh apparatov* (*Transportnyye sistemy*) (Moscow: Mashinostroyeniye, 1985).

54. Mikhail Rebrov, "The Last Argument: A Study of the Designer in Black and White" (English title), *Krasnaya zvezda*, March 25, 1995, p. 6.

55. For the émigré, see Victor Yevsikov, *Re-Entry Technology and the Soviet Space Program (Some Personal Observations)* (Falls Church, VA: Delphic Associates, 1982). For the French journalist, see C. Wachtel, "The Chief Designers of the Soviet Space Program," *Journal of the British Interplanetary Society* 38 (December 1985): 551-62.

program, timed to compete with Apollo.⁵⁸ The following month, veteran cosmonaut Valeriy F. Bykovskiy, one of those who had trained for the project, admitted the same thing in his just-published biography.⁵⁹ These two publications burst the floodgates. Within weeks, there was a major in-depth article in *Izvestiya* on the NI program, and by October 1989, Mishin gave a long interview on the project, covering his role as one of the pioneers of the Soviet space program.⁶⁰

In this and subsequent interviews, Mishin did not hide his bitterness at having been wiped from the history of the Soviet space program. He took aim at Glushko, Ustinov, and all the other individuals who had silenced him for fifteen years. He also did not have kind words for the current Soviet space program:

*Very little has been done about what we thought about and dreamed about 20 years ago, even 30 years ago with Korolev. It is simply vexing that so few useful and efficient space vehicles are in Earth orbit . . . we have become addicted to the long, monotonous long-duration manned missions in the tight Salyut-Mir which repeat each other. It is very wasteful.*⁶¹

When asked how there could be a vigorous forward-looking space program, he replied:

*Space exploration has been hampered by monopoly and secrecy, and by nepotism and politically dealing in the allocation of buildings and subsidies. We need broad, open competition in projects for a unified technical task. And discussion of tasks, ideas, and proposals, and independent expert evaluations, and open selection of the winners. Only after this, in full view of everyone, should there be implementation of projects in which the whole of society is convinced of their need and soundness.*⁶²

He might as well have been talking of an alien world as compared to the Soviet system. Mishin also added his own two cents to the emerging debate over whether if Korolev had lived, the Soviets might not have had more success in their ventures into space:

*. . . in the main thing, in the desire to create a well-considered strategy for space exploration, we were, I hope, fellow thinkers. No, I probably did not possess the kind of will and sharp tongue that distinguished Korolev. I am prepared to admit that. But in our space situation, the replacement of one character for another and the replacement of leading personalities did not play any decisive role.*⁶³

In the initial flurry of publicity concerning the Soviet Union's aborted piloted lunar program, Mishin wrote extensively and granted many interviews, but in recent years, he has remained out of the public eye, except to occasionally author pieces paying tribute to his mentor, Sergey P. Korolev. The latter clearly had a very high regard for Mishin, having picked the thirty-year-old Mishin to serve as principal deputy in 1946. Korolev told a journalist once:

I found this to be true and recurring regardless of circumstances: every now and again, all of a sudden a man would come from out of nowhere, from the great unknown, a

58. Lev Kamanin, "From the Earth to the Moon and Back" (English title), *Poisk* 12 (July 1989): 7-8.

59. Grigoriy Reznichenko, *Kosmonavt-5* (Moscow: Politicheskoy literatury, 1989), p. 98.

60. Sergey Leskov, "How We Didn't Get to the Moon" (English title), *Izvestiya*, August 18, 1989, p. 3; A. Tarasov, "Missions in Dreams and Reality" (English title), *Pravda*, October 20, 1989, p. 4.

*man that would be remarkable precisely for the qualities you sought: he is gifted, courageous, honest. . . . He would introduce himself, extend his hand in a trustworthy manner, modestly speak of the work he has done, and a miracle would happen, the unknown is no longer the unknown. And then you would say to yourself: "This is he, precisely the man I need."*⁶⁴

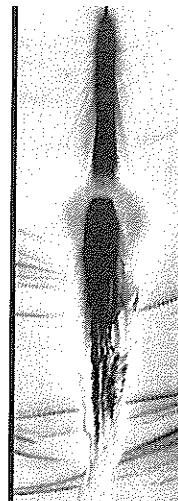
He was evidently speaking of Mishin.

Perhaps wanting his story to be told, Mishin put his personal diaries—thirty-one volumes covering the period from 1960 to 1974—up for sale at a special auction of Soviet and Russian space artifacts at Sotheby's in late 1993.⁶⁵ The Perot Foundation purchased the diaries for \$190,000. Unfortunately at the time of this writing, the institution has yet to make these priceless writings available to scholars. Portions have been exhibited as part of the National Air and Space Museum's "Space Race" exhibit opened in 1997. Mishin himself continues to teach at the Moscow Aviation Institute, having just turned eighty-two. He wrote perhaps the best epitaph to his own contribution to the Soviet piloted lunar program in a monograph he authored in 1990 on the N1-L3 program:

*I do not want the readers to think that I am trying to relieve myself as Chief Designer of responsibility for certain errors committed (including by me personally) during work on the lunar program. Only he who does nothing makes no mistakes. We, the successors of S. P. Korolev, did everything we could, but our efforts proved to be inadequate.*⁶⁶

64. Rebrov, "The Last Argument."

65. *Russian Space History, Sale 6516* (New York: Sotheby's, 1993), description for Lot 29: Iohn Noble



CHAPTER TWENTY CODA

For a brief period in the late 1950s and early 1960s, one could reasonably argue that the Soviet Union was the leading spacefaring nation in the world. In the light of the ultimate demise of the Soviet empire, however, thinking of the USSR as launching humanity's first steps into the cosmos seems a strange abstraction—the memory feels oddly empty, almost irrelevant perhaps. When we do remember, we tend to divorce Sputnik from its origin as a uniquely Soviet artifact—an eighty-four-kilogram sphere that was designed and built by men who lived through a war in which their country lost more than 25 million people, experienced the terror of Stalinist times, and defined themselves as Communists. Instead, we focus overwhelmingly on the *impact* of Sputnik rather than the construction of the artifact itself. I do not mean to suggest that meanings are unimportant. But in privileging only Sputnik's impact, we have told only half the story. Certainly, this is partly because the Soviet space program was given birth—and given flight—behind closed doors. Peeking through the now opened curtains, what I have tried to present here is an account of the missing half of that tale. This is only one version of the story, and certainly not the only one. But in sifting through the evidence and constructing the narrative, three broad themes have served as guidelines.

The first theme has to do with the institutional framework and the interplay among different factions within the Soviet space program and its antecedent missile project. Four primary constituencies were fundamental to establishing a Soviet ballistic missile program in the 1940s and 1950s. They were the engineers, the artillery officers, the defense industrialists, and the Communist Party. Each faction had its own agenda, but for a period of about fifteen years following the end of the war, their motivations intersected at crucial points to give rise to the world's first space program. The engineers were driven by their somewhat idealistic dreams of exploring space—dreams that had taken flight when they were young rocketry enthusiasts in the 1930s. The artillery officers were in need of a new generation of strategic weaponry to transform their backdated service into a powerful deterrent force. The defense industrialists had the not inconsiderable task of expediting the development of a strong military. And the Party leaders—in particular Stalin and Khrushchev—were driven by the political exigencies of the Cold War to direct the three other factions to elevate the Soviet Union from a nation afraid to defend itself to one that could threaten with offense.

The collusion of these four groups was necessary for the development of the world's first intercontinental ballistic missile, the R-7. This missile, of course, was simply a military weapon—a tool for mass destruction. In the hands of one visionary—Sergey Pavlovich Korolev—it became something entirely different. In the unlikely marriage of military imperative

country that had been dismissed as a nation of farmers and factory workers had suddenly arrived on the world's stage with an achievement that was too impressive to ignore.

There was more to come. Within four years, using the same rocket that had launched Sputnik, the Soviet Union, now armed with a new tint to the old socialist doctrine of harnessing technologies in the interest of the state, reached the apotheosis of its dizzying trajectory into the heavens. Historian Walter A. McDougall, writing in the introduction of his seminal work . . . *the Heavens and the Earth*, compared the event to the migration of the fish Eusthenopteron from the seas to the land:

In A.D. 1961 Homo sapiens, in turn, left the realm of solids and gases and lived, for 108 minutes, in outer space. Life again escaped, or by definition extended, the biosphere. The earth's crust and canopy of air became another platform to a new universe as infinite as soil and sky must have seemed to Eusthenopteron.'

Only the vicissitudes of history will decide whether the flight of Yuriy Alekseyevich Gagarin in the spaceship Vostok will be remembered with such sweeping comparisons in the centuries hence. Even as the decade of the 1960s passed through tumult and chaos, humankind's first trip into space began to recede into the background. By the time that the first humans landed on the surface of the Moon in 1969, Gagarin's flight had been eclipsed in the popular conception of space exploration by the spectral images of two American men who left their footprints on another planetary body. The technology, the men, the pictures, and even the parades seemed so much more compelling to a new generation. In the historiography of space exploration, Gagarin's excursion assumed more importance for how it affected the American decision to aim for the Moon than for its own place in the history of human evolution.

But Gagarin's flight, both from a historical and a technological perspective, warrants more scrutiny. This is not only because it was achieved by a nation that was not expected to do so, but, simply and ultimately, because it was, as McDougall pointed out, an event that, like perhaps Apollo 11, transcended nations, languages, cultures, and continents and, for 108 minutes, represented the planet Earth: for the first time since the origin of life on this planet, one life form had managed to escape it. At the same time, we should not minimize more earthly considerations. Gagarin's flight did not, after all, happen in isolation from the political, economic, and social dimensions of the Cold War. And ironically, as this book has shown, the same forces that allowed the Soviet Union to send the first human into space—the need to arm themselves with powerful new weapons—deprived the country of further national triumphs in the space race.

Considering the post-Gagarin era leads to the second major theme of this work: the Soviet effort to beat the United States in landing the first person on the Moon. After an unprecedented catalogue of firsts in the late 1950s and early 1960s, the Soviets failed dismally in this quest. The road to failure began almost as soon as Gagarin had floated down in his parachute. After 1961, the Soviet space program, jostling for a role in the new Soviet military technocracy, began to stumble and slide in trying to attain a stable position of growth. Different factions were all vying for the same piece of pie. The artillery officers, now subsumed under the Strategic Missile Forces, increasingly declined to fund the primarily civilian endeavors of the human space program. The military, it seems, was more interested in missiles than the Moon. Grand visions of space exploration, as the one Korolev proposed in 1960, died under their own weight as the military siphoned off funding from important space projects in favor of developing a new generation of strategic weapons systems. Because their primary job was to design intercontinental

ballistic missiles, the main space design organizations had to sacrifice ambitious space plans on the altar of strategic necessity. The Cold War, having given birth to the Soviet space program, would seriously threaten its very existence. In this climate, important avenues of research, such as the development of high-energy cryogenic rocket engine technology, never reached beyond the exploratory stage.

Despite the visible firebrand rhetoric of Nikita S. Khrushchev in the early 1960s, his support of a coherent long-range civilian space program was lukewarm at best. In 1961, when U.S. President John F. Kennedy laid down the challenge of reaching the surface of the Moon prior to the end of the decade, the Soviet space leadership hardly took notice. Organizational chaos emerged as a flurry of competitors began to dilute the hard-earned gains of the space program. The engineer faction, so united at the time of Sputnik, began to fragment in the face of limited funding. Between 1961 and 1964, Korolev ran his program on a shoestring budget, as proposal after proposal ended up in ministry file cabinets, never to be seen again. In desperation, he mounted two hastily prepared spectacles in the mid-1960s—missions that had no value other than to please the Party leaders: the launch of the first multicosmonaut crew in 1964 and the accomplishment of the first spacewalk in 1965. The diversion cost the Soviets dearly. It was only in 1964 that Khrushchev sanctioned a piloted lunar landing program, three years after Kennedy's own challenge. The commitment itself was never followed up as the military continued to withhold funding, prompting the engineers to omit crucial phases in the ground testing of their lunar rocket. The shortcuts inexorably led to the series of crushing failures just as the United States was landing its first citizens on the surface of the Moon.

With the loss of the Moon race in 1969, the Soviets diverted much of their resources in the following years to space station programs. Korolev's successor, Vasiliy Mishin, has argued in recent years that despite the success of Apollo, there was no cause to abandon the massive N1-L3 lunar program simply because the Americans had arrived at the Moon first. As sound as this logic seems to be, engineers in the 1970s had to deal with a political climate that was vehemently hostile to expensive civilian programs such as lunar missions. Soviet leaders saw little need for such projects, because their success would raise inevitable questions about the original failure to beat Apollo.

The third and final theme of this work addresses the manner in which the Soviets handled technological innovation in the space program. The evidence both confirms and counters our *a priori* conceptions of Soviet technology as one characterized by evolutionary rather than revolutionary changes. In the space program, the Soviets used a combination of both; the decision to forego the former in favor of the latter often had as much to do with accident as with political expediency. For example, having built the first piloted spaceship Vostok by 1960, the Soviets tried hard to extend its capabilities by introducing relatively minor changes that cumulatively added to moderate gains. They abandoned Vostok as a viable piloted spaceship only in 1966 when they absolutely had to—that is, when the Soyuz spacecraft, which represented a qualitative leap in design and performance, was virtually ready. The decision to fly one over the other in 1966 had as much to do with the impossibility of fulfilling contemporary objectives in space with the Vostok as with the fact that flying the Vostok (or its surrogate Voskhod) in 1966 would have demonstrated a visible and obvious lag to U.S. space technology.

In the thirty-year period spanning from 1945 to the mid-1970s, I looked at two cases of technological leaps: the R-7 ICBM and the N1 Moon rocket. Both projects required immense coordination in scale and scope across a Byzantine state-controlled landscape of research, development, and production. The success of the R-7 resulted not only from the high degree of financial commitment afforded by the state, but also because of the use of unorthodox man-

could substantiate the proposal and ultimately convince the Soviet government of its feasibility. The council also served as an unusual managerial mechanism that allowed chief designers to intervene at key points in the development of the ICBM. Ultimately, the Council of Chief Designers managed to circumvent the internal self-generated inertia of Soviet industry, which discouraged major technological leaps and favored short-term gains.

In his important study on the origins of the Soviet atomic bomb, *Stalin and the Bomb*, David Holloway concluded that "[a]fter Stalin's death, nuclear scientists . . . enjoyed unprecedented authority among the political leaders."² The evidence from the space program suggests that the privilege of authority granted to the nuclear scientists was eventually expanded to include the engineers who played influential roles in the rise of the Soviet ballistic missile program. Although Western observers have long thought that it was Sputnik that changed the fortunes of these engineers such as Korolev and Glushko, their relationship to the political leadership changed more than a year before Sputnik with the successful test of the first Soviet strategic missile, the R-5M. The landmark test dramatically escalated the space engineers' leverage with both the Communist Party and the government and eventually led to the formation of a loose coalition of designers who would wield considerable power and influence. Although after 1960 they rarely, if ever, acted as a united front, the missile and space designers represented a formidable constituency that profoundly affected the direction of space policy beginning the late 1950s.

Because the Soviet political leadership lacked a clear understanding of the new technologies of the missile and space program, they needed the engineers to actively participate in policy formulation. The chief designers obliged willingly by forming lobbies, and, in the process, they acquired sufficient power to oppose important mandates from the top. There should be no confusion as to how the designers attained their powerful positions—it was not space, but rather their contribution to missile development that empowered them. Because the space program was largely a byproduct of missile production, the privileges almost by default were extended from the latter to the former. Both Nikita Khrushchev and Sergey Korolev played key roles in this process: Khrushchev because he allowed the rise of a constituency, and Korolev because he strongly pushed for it. But as the powerful chief designers vied for limited resources, they began to abuse the patronage system through various contacts within the Central Committee. "The favor of not even Khrushchev, Brezhnev, or Ustinov, but of a totally forgotten Central Committee agent," one Russian journalist wrote, "could determine the prospects for the development of the highly complex [space] sector for years."³ The chaos was one of the key factors in the failure of the N1 program.

Korolev did not live long enough to witness the ultimate decline of the juggernaut he helped create. He did, however, leave behind an unmatched legacy—one that continues to be debated more than thirty years after his death. Most historians, both in Russia and in the West, have not argued with the holy grail of the history of the Soviet space program: that Sergey Pavlovich Korolev was its founder and central motivator. Given what is known about the vortex of events surrounding the launch of both Sputnik and Gagarin, it would be hard to dispute that claim. But at the same time, there has been an eagerness to attribute to Korolev roles that he clearly did not play, at least in his later life. It is particularly noteworthy that Sergey Korolev, the person who was most responsible for Sputnik was neither a scientist nor an engineer, but rather a manager with a vision. Boris V. Raushenbakh, one of Korolev's close associates from the 1960s, later wrote:

Sometimes one hears it said that Korolyov was an excellent engineer and scientist. It is difficult to agree with this if the terms "engineer" and "scientist" are accorded their usual meaning. Korolyov himself did not devise any especially interesting solution to a complicated structural problem, as is the case with brilliant engineers. He was also not a scientist in the usual sense of the word: his name is not linked with any original scientific theory or with any prolonged and extensive study of a complicated phenomenon. However, these statements are not to be construed as a deprecation of the role which he played in the birth of space travel. There are many outstanding scientists and engineers, and Korolyov is a unique individual. His uniqueness, moreover, is linked to the fact that he was to introduce a new era into human history: the space age.⁴

The rise of the Soviet space program was one of the most significant processes in the history of science and technology in this century, not only because it opened what Raushenbakh calls "the space age," but also because it had profound sociopolitical consequences all over the world. Within the context of the Cold War itself, the Soviet space effort was a benchmark—a milestone that turned history from one path to another. For the first decade after Sputnik, the space age was indistinguishable in many ways from the space race. As the breadth of retrospect grows longer and longer, the import of the latter—that is, the space race—will no doubt recede far into the background, as perhaps it should. But for a short period in this century, the race provided the impulse for humankind to depart from this planet and reach the Moon. The Soviets, of course, lost this race, although they managed to throw shreds of doubt onto the victor's parade by denying that they had even signed up for the event. This deception existed for more than two decades, and when the truth was finally revealed, few took notice.

In 1999, during the thirtieth anniversary of the landing of humans on the surface of the Moon, the memory of Apollo spurred a brief but important resurgence of the sense of wonder and fascination that humans attach to space exploration. But lost amid the reevaluations and archaeology digs through Apollo, perhaps the greatest technological adventure of humankind was the other side of the coin—the story of those who had given reason to embark on Apollo in the first place. Buried under history was Korolev's "last love," the N1 program. That the N1 program was consigned to the status of a footnote is not so unusual: history has a way of privileging successes over failures. Our understanding of this dichotomy, between success and failure, is intrinsically tied to a second one—that between inevitability and contingency. On the one hand, we tend to see an inevitability in history's trajectory to the present—for example, that given the set of prevailing circumstances, the N1 program had to fail and that Apollo had to succeed. On the other hand, the story is compelling precisely because the outcome was not inevitable—that is, it was contingent on thousands of circumstantial factors. The tension between contingency and inevitability has contributed much to the enduring myths we now associate with the race to the Moon. The story of the Soviet space program has long been part of that myth. We have tended to see Soviet successes in space (such as Sputnik) as contingent and Soviet failures (such as the N1) as inevitable. This myth, it seems, is far too simplistic and takes away from the genuinely worthy accomplishments of Sputnik and Gagarin. The myth served its purpose during the Cold War, but now with the collapse of the Soviet Union and the opening of the Russian archives, it is finally time to put it away.

2. David Holloway, *Stalin and the Bomb: The Soviet Union and Atomic Energy, 1939–1956* (New Haven,